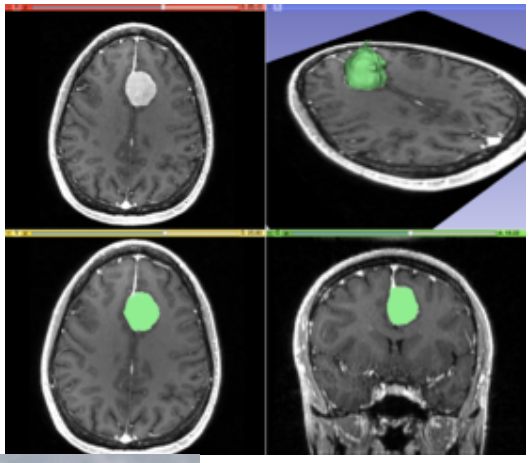
# Computer Vision

## "Data Science Applications in Physics" *Balkan School in Tirana 2024*

### What is Computer Vision (CV)?

Is interdisciplinary field of AI that:

- Focuses on enabling computers to acquire, process, analyze, and understand digital images and videos.
- Develops algorithms that gives computers high-level understanding from visual data (digital images & videos) to produce numerical or symbolic information.
- Seeks to automate tasks that the human visual system can do.

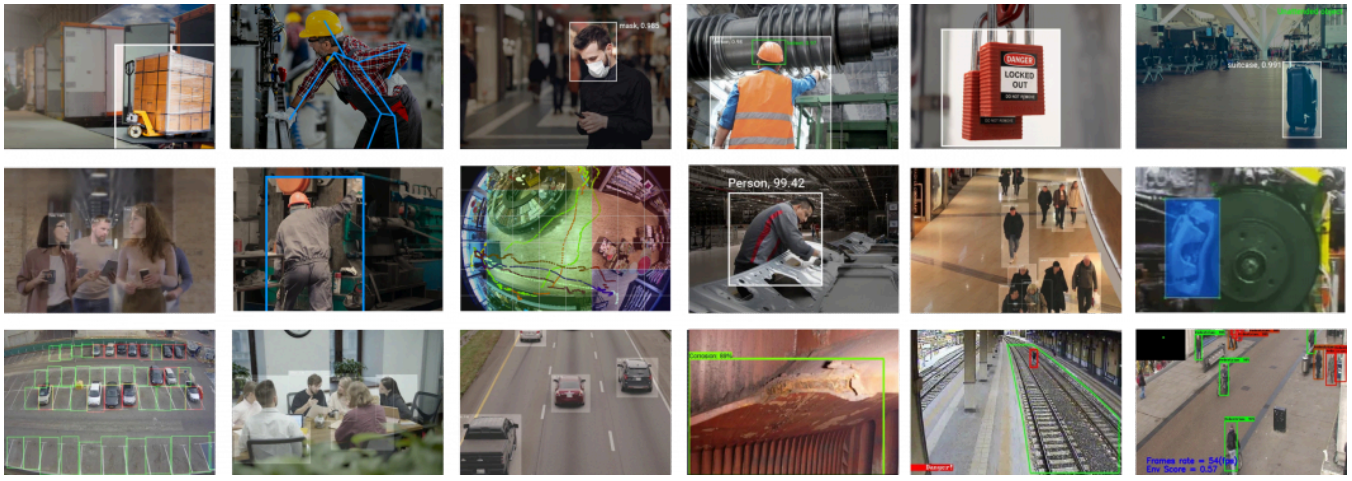


### What are the tasks of Computer Vision?

- Image classification,
- object detection,
- image segmentation,
- facial recognition,
- scene understanding, etc...

### Computer Vision Applications

- Self-driving cars,
- surveillance systems,
- medical imaging,
- augmented reality,
- robotics
- quality control in manufacturing
- **PHYSICS Experiments**




# Requirements!

Here we will use python API of two popular libraries:

1. **OpenCV** (https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html)
2. **YOLOv8** from ultralytics (https://docs.ultralytics.com/)

** Prerequisites: Python & numpy **
For quick introduction to Python see e.g. https://python.swaroopch.com/ or https://www.freecodecamp.org/news/the-python-guide-for-beginners/

For a quick start with **Numpy** see e.g. https://numpy.org/doc/stable/user/quickstart.html

## Installation

Assuming you have a conda environment up and running you should create a new environment by running the following command in the terminal

```
conda create --name cv
```

then activate the new environment

```
conda activate cv
```

then install **OpenCV**

```
conda install -c conda-forge opencv
```

Install **Ultralytics** by excecuting the following command in the terminal:

```
pip install ultralytics

conda install pytorch torchvision ultralytics
```

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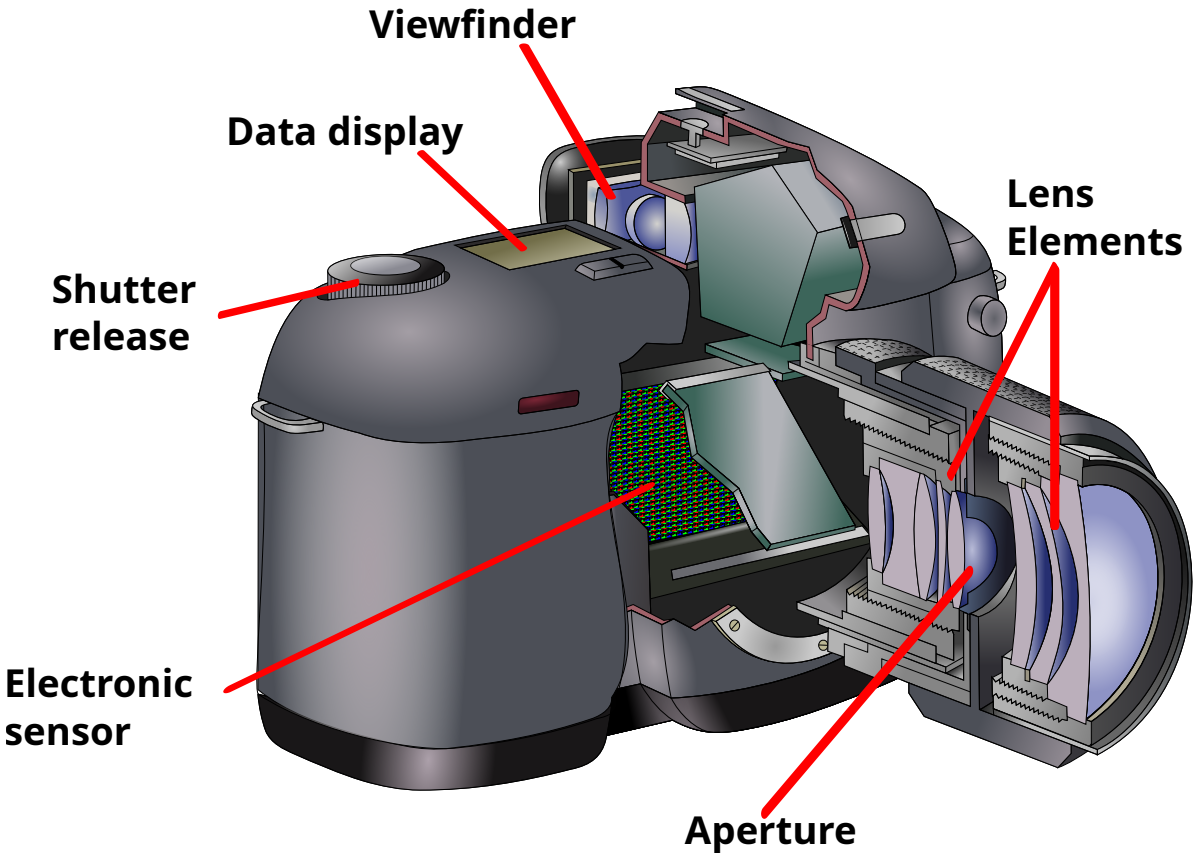
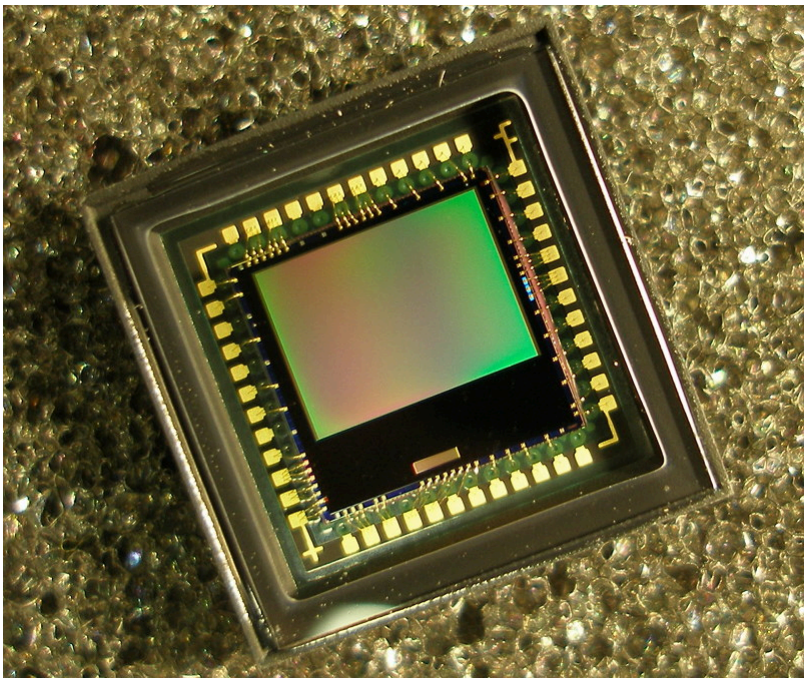
# Camera and its Sensor

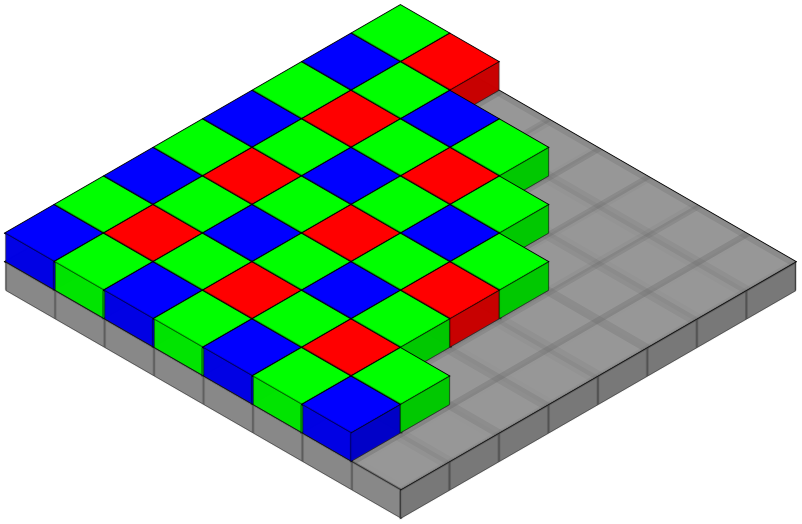
A typical digital camera has an **optical system** an **image sensor** a **data aquisition circuit and a digital image processing unit** (+ a display).



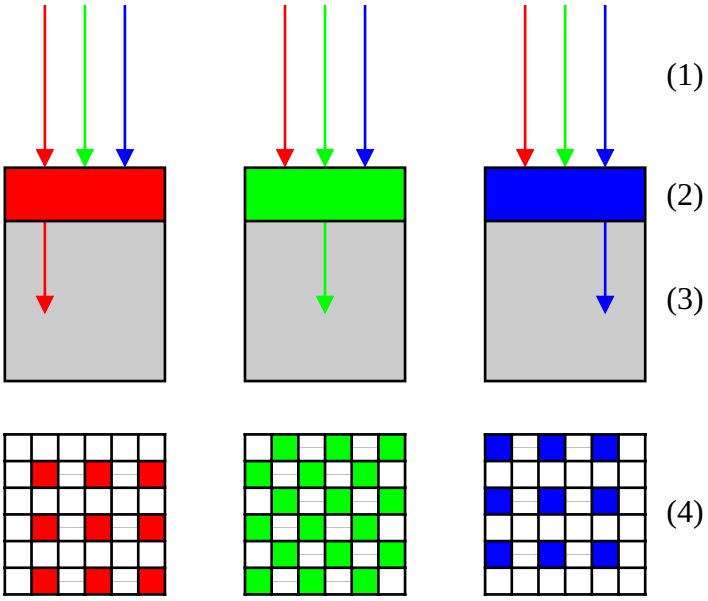
This is how an image sensor (camera sensor) looks like



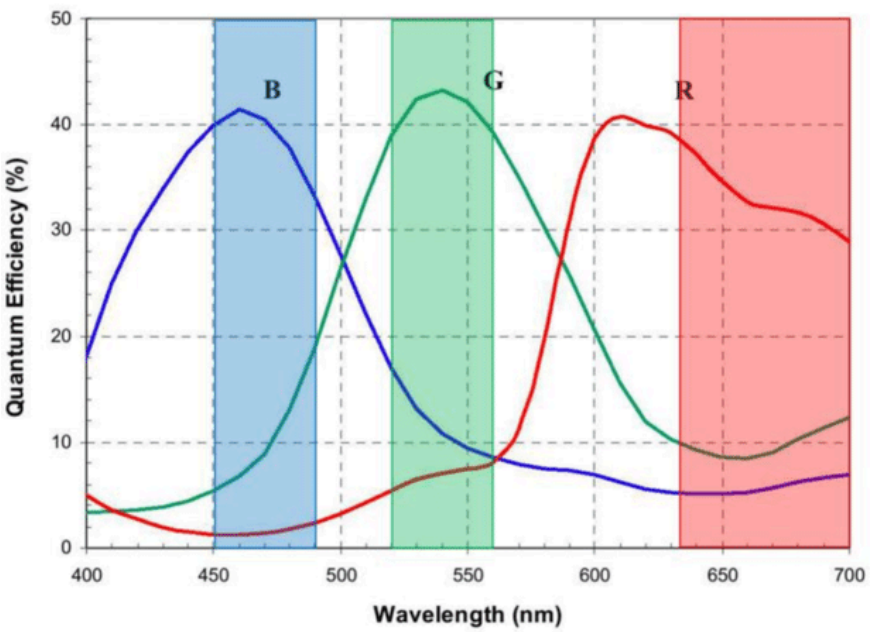
The pixels sensitive to different colors have different arrangements. The most common is the Bayer pattern:



Pixels are semiconductors (photo-sensitive p-n junctions)




Typical pixel quantum efficiency (sensitive spectrum)




| | |
|---|---|
| Red | ~ 700–635 nm |
| Green | ~ 560–520 nm |
| Blue | ~ 490–450 nm |

# Demonstration

## Load an Image

```
import cv2
img = cv2.imread("DATA/buss.jpg") # loads the image in BGR color scheem
cv2.imshow("Display window", img) # shows the image in a dedicated canvas
```

An image is nothing but a matrix of numbers with values 0-255 (8-bit color depth). The dimensions of this matrix are the resolution of the image $N \times M \times 3$ for three colors Red (R), Green (G) and Blue (B).

```
import cv2
img = cv2.imread("DATA/bus.jpg")
#cv2.imshow("Display window", img)
img.shape
print(f'Showing first 10x10 pixels of the image for each color\n\nBlue=
{img[:10,:10,0]}\n\n\nGreen={img[:10,:10,1]}\n\n\nRed={img[:10,:10,2]}')
```

---

## Load Videos or Cameras

We can show video from camera, loaded from disk or from internet e.g. from youtube. Check the following example.

```
import cv2
cap = cv.VideoCapture(0) #camera 0 is the first, 1 is second etc... Changes this to select the camera you want to connect.
while True:
    # Capture frame-by-frame
    ret, frame = cap.read()
    # Display the resulting frame
    cv.imshow('frame', frame)
    if cv.waitKey(1) == ord('q'):
        break
# When everything done, release the capture
cap.release()
cv.destroyAllWindows()
```

If you want to change the image into Gray:

```
import cv2
cap = cv.VideoCapture(0) #camera 0 is the first, 1 is second etc... Changes this to select the camera you want to connect.
while True:
    # Capture frame-by-frame
    ret, frame = cap.read()
    # Our operations on the frame come here
    gray = cv.cvtColor(frame, cv.COLOR_BGR2GRAY)
    # Display the resulting frame
    cv.imshow('frame', gray)
    if cv.waitKey(1) == ord('q'):
        break
# When everything done, release the capture
cap.release()
cv.destroyAllWindows()
```

```
import numpy as np
import cv2
cap = cv2.VideoCapture(0)
if not cap.isOpened():
    print("Cannot open camera")
    exit()
while True:
    # Capture frame-by-frame
    ret, frame = cap.read()
    # if frame is read correctly ret is True
    if not ret:
        print("Can't receive frame (stream end?). Exiting ...")
        break
    # Display the resulting frame
    # cv2.imshow('frame', frame) # show only the Blue part
    # cv2.imshow('frame', frame[:,:,0]) # show only the Blue part
    # cv2.imshow('frame', frame[:,:,1]) # show only the Green part
    cv2.imshow('frame', frame[:,:,2]) # show only the Red part
    if cv2.waitKey(1) == ord('q'):
```

```
        break
# When everything done, release the capture
cap.release()
cv2.destroyAllWindows()
```

---

# Controlling camera properties

We can controll camera settings via OpenCV. If you want to show camera settings in linux you can use e.g. `v4l2-ctl` (see `man v4l2-ctl` for help). Here we are interested in changing the frame rate, thus use:

```
**v4l2-ctl --list-formats-ext -d /dev/video0**
```

change 0 in `/dev/video0` in to whatever camera you are using, e.g. `/dev/video2` etc. The result of the command for the camera I am using here is:

## [0]: 'MJPG' (Motion-JPEG, compressed)

```
    **1920x1080 (30.000 fps)**
    **1280x720  (60.000 fps)**
    **1024x768  (30.000 fps)**
    **640x480 (120.101 fps)**
    **800x600  (60.000 fps)**
    **1280x1024 (30.000 fps)**
    **320x240 (120.101 fps)**
```

## [1]: 'YUYV' (YUYV 4:2:2)

```
    **1920x1080 (6.000 fps)**
    **1280x720  (9.000 fps)**
    **1024x768  (6.000 fps)**
    ** 640x480 (30.000 fps)**
    ** 800x600 (20.000 fps)**
    **1280x1024 (6.000 fps)**
    * *320x240 (30.000 fps)**
```

Therefore we can use one of these options, e.g. let us make a 6 fps

```
import cv2
cap = cv2.VideoCapture(0)

# cap.set(cv2.CAP_PROP_FOURCC, cv2.VideoWriter_fourcc('M','J','P','G'))
# cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
# cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)
# cap.set(cv2.CAP_PROP_FPS, 60)

cap.set(cv2.CAP_PROP_FOURCC, cv2.VideoWriter_fourcc('Y','U','Y','V'))
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1024)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 768)
cap.set(cv2.CAP_PROP_FPS, 6)
```

```
while True:
    # Capture frame-by-frame
    ret, frame = cap.read()
    cv2.imshow('frame', frame)
    if cv2.waitKey(1) == ord('q'):
        break
# When everything done, release the capture
cap.release()
cv2.destroyAllWindows()
```

### Load video from source

Using some video file you can load it directly

```
import cv2
cap = cv2.VideoCapture("DATA/traffic.mp4")
if not cap.isOpened():
    print("Cannot open camera")
    exit()

while True:
    ret, frame = cap.read()
    if not ret:
        print("Can't receive frame (stream end?). Exiting ...")
        break
    cv2.imshow('frame', frame) # show only the Blue part
    if cv2.waitKey(1) == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

---

## Image processing

If you are interested in image processing (a very important topic) you have the following tutorials:

- **OpenCV**: https://docs.opencv.org/4.x/d2/d96/tutorial_py_table_of_contents_imgproc.html
- **scikit-image**: https://scikit-image.org/docs/dev/auto_examples/

Here you can every image manipilation such as:

- Changing Colorspace
- Geometry Transformations
- Image Thresholding
- Smoothing Images
- Morphological Transformations
- Image Gradients
- Edge Detection

- Contours & Histograms
- Template Matching
- Hough Line & Circle Transforms
- Image Segmentation
- etc ...

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## Object Detection and Tracking with Neural Networks

A simple code how to use YOLOv8

```
from ultralytics import YOLO

# Load model
model = YOLO('yolov8n.pt')  # Load an official Detect model
#model = YOLO('yolov8n-seg.pt')  # Load an official Segment model
#model = YOLO('yolov8n-pose.pt')  # Load an official Pose model

src = "DATA/airplain.jpg"

results = model.track(source=src, show=True)  # Tracking with default tracker
#results = model.track(source=src, show=True, tracker="bytetrack.yaml")  # Tracking with
ByteTrack tracker
```

```
import cv2
from ultralytics import YOLO

# Load the YOLOv8 model
model = YOLO('yolov8n.pt')  # Load an official Detect model
#model = YOLO('yolov8n-seg.pt')  # Load an official Segment model
#model = YOLO('yolov8n-pose.pt')  # Load an official Pose model


#src = "https://www.youtube.com/watch?v=xRdQmO9qLHo"
#src = "DATA/dance.mp4"
src = "DATA/traffic.mp4"
#src = "DATA/peopleWalking.mp4"
#src = "DATA/airplain.jpg"

cap = cv2.VideoCapture(src)

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
```

```
        # Run YOLOv8 tracking on the frame, persisting tracks between frames
        #results = model.track(frame, persist=True, tracker="botsort.yaml", conf=0.3,
iou=0.5)
        results = model.track(frame, persist=True, tracker="bytetrack.yaml", conf=0.6,
iou=0.5)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Display the annotated frame
        cv2.imshow("YOLOv8 Tracking", annotated_frame)
        #cv2.imshow("YOLOv8 Tracking", frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break

# Release the video capture object and close the display window
cap.release()
cv2.destroyAllWindows()
```

## Use camera to detect objects from pretrained categories

Use pretrained model classes of objects to detect.

```
from ultralytics import YOLO
import cv2
import math
# start webcam
cap = cv2.VideoCapture(0)
cap.set(3, 640)
cap.set(4, 480)

# model
model = YOLO("yolo-Weights/yolov8n.pt")

# object classes
classNames = ["person", "bicycle", "car", "motorbike", "aeroplane", "bus", "train", "truck",
"boat",
              "traffic light", "fire hydrant", "stop sign", "parking meter", "bench",
"bird", "cat",
              "dog", "horse", "sheep", "cow", "elephant", "bear", "zebra", "giraffe",
"backpack", "umbrella",
              "handbag", "tie", "suitcase", "frisbee", "skis", "snowboard", "sports ball",
"kite", "baseball bat",
              "baseball glove", "skateboard", "surfboard", "tennis racket", "bottle", "wine
glass", "cup",
```

```
              "fork", "knife", "spoon", "bowl", "banana", "apple", "sandwich", "orange",
"broccoli",
              "carrot", "hot dog", "pizza", "donut", "cake", "chair", "sofa", "pottedplant",
"bed",
              "diningtable", "toilet", "tvmonitor", "laptop", "mouse", "remote", "keyboard",
"cell phone",
              "microwave", "oven", "toaster", "sink", "refrigerator", "book", "clock",
"vase", "scissors",
              "teddy bear", "hair drier", "toothbrush"
              ]


while True:
    success, img = cap.read()
    results = model(img, stream=True)

    # coordinates
    for r in results:
        boxes = r.boxes

        for box in boxes:
            # bounding box
            x1, y1, x2, y2 = box.xyxy[0]
            x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2) # convert to int values

            # put box in cam
            cv2.rectangle(img, (x1, y1), (x2, y2), (255, 0, 255), 3)

            # confidence
            confidence = math.ceil((box.conf[0]*100))/100
            print("Confidence --->",confidence)

            # class name
            cls = int(box.cls[0])
            print("Class name -->", classNames[cls])

            # object details
            org = [x1, y1]
            font = cv2.FONT_HERSHEY_SIMPLEX
            fontScale = 1
            color = (255, 100, 0)
            thickness = 2

            cv2.putText(img, classNames[cls], org, font, fontScale, color, thickness)

    cv2.imshow('Webcam', img)
    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

# Train and Validate and Load Your Model

This is how you load a model, train it, evaluate its performance on a validation set, and even export it to ONNX format.

```
from ultralytics import YOLO

# Create a new YOLO model from scratch
model = YOLO('yolov8n.yaml')

# Load a pretrained YOLO model (recommended for training)
model = YOLO('yolov8n.pt')

# Train the model using the 'coco128.yaml' dataset for 3 epochs
results = model.train(data='coco128.yaml', epochs=3)

# Evaluate the model's performance on the validation set
results = model.val()

# Perform object detection on an image using the model
results = model('https://ultralytics.com/images/bus.jpg')

# Export the model to ONNX format
success = model.export(format='onnx')
```